

TECHNICAL SPECIFICATION DOCUMENT

Smart Waste Sorting Terminal - "EcoVision Terminal"

ARCHITECTURAL DECISION LOG

Decision 1: AI Inference Strategy

Alternative A: Cloud-based Classification

- **Pros**: Unlimited model complexity, continuous learning, centralized updates
- **Cons**: Requires constant connectivity (violates offline-first), 3-5s latency, monthly API costs
- **Cost**: \$15-30/month/terminal
- **Inference Speed**: 3-5 seconds (network + processing)
- **UX Impact**: Frustrating delays, failure in connectivity outages
- **Durability**: Dependent on external services

Alternative B: Edge AI with TensorFlow Lite

- **Pros**: 100% offline operation, <500ms inference, no recurring costs
- **Cons**: Limited model complexity, manual update deployment
- **Cost**: One-time development only
- **Inference Speed**: 300-800ms on RPi4
- **UX Impact**: Real-time feedback, consistent performance
- **Durability**: No external dependencies

Alternative C: Hybrid (Edge Primary + Cloud Enhancement)

- **Pros**: Offline reliability with optional cloud learning
- **Cons**: Increased complexity, dual maintenance burden
- **Cost**: Development overhead + optional cloud costs
- **Inference Speed**: 300-800ms (edge) or fallback to cloud
- **UX Impact**: Best of both worlds but complex
- **Durability**: High with proper fallback logic

DECISION: **Alternative B (Edge AI with TensorFlow Lite)**

- **Reasoning**: Offline-first constraint is non-negotiable. 10-second interaction budget allows for 300-800ms inference. RPi4 handles TFLite models efficiently. Municipal budget likely prefers CAPEX over OPEX.

Decision 2: Computer Vision Model Architecture

Alternative A: Custom CNN (trained from scratch)

- **Pros**: Optimized for specific waste categories
- **Cons**: Requires 50k+ labeled images, 4-6 weeks training time, expertise needed
- **Cost**: High (dataset creation + GPU training time)
- **Inference Speed**: ~400ms (optimized)
- **UX Impact**: Best accuracy potential
- **Durability**: Requires ML expertise for updates

Alternative B: Transfer Learning (MobileNetV3 base)

- **Pros**: Requires only 2-5k labeled images, 3-5 days fine-tuning, proven architecture
- **Cons**: Slightly larger model size than custom
- **Cost**: Moderate (dataset labeling + training)
- **Inference Speed**: ~600ms on RPi4
- **UX Impact**: 85-92% accuracy achievable

- **Durability**: Industry-standard, well-documented

Alternative C: Pre-trained General Object Detection (YOLO/SSD)

- **Pros**: Zero training time, immediate deployment
- **Cons**: Not optimized for waste classification, poor accuracy for packaging nuances
- **Cost**: Minimal
- **Inference Speed**: ~800ms
- **UX Impact**: 60-70% accuracy (insufficient)
- **Durability**: Limited improvement path

DECISION: **Alternative B (MobileNetV3 Transfer Learning)**

- **Reasoning**: 72-hour prototype constraint requires rapid development. MobileNetV3 optimized for mobile/edge devices. Transfer learning balances speed-to-deployment with accuracy. Community has pre-existing waste classification datasets (TrashNet, Waste Classification Data V2).

Decision 3: UI Framework

Alternative A: Native Python (Tkinter/Kivy)

- **Pros**: Direct hardware access, no browser overhead
- **Cons**: Limited LLM code generation quality, primitive UI components
- **Cost**: Low
- **Inference Speed**: N/A (UI framework)
- **UX Impact**: Dated appearance, harder animations
- **Durability**: Stable but limited ecosystem

Alternative B: Electron-based Web Stack

- **Pros**: Rich UI libraries, excellent LLM training data (HTML/CSS/JS)
- **Cons**: Heavy memory footprint (1-1.5GB), slower startup
- **Cost**: Moderate (resource overhead)
- **Inference Speed**: N/A
- **UX Impact**: Modern, fluid animations
- **Durability**: Active ecosystem

Alternative C: Web Stack with Lightweight Browser (Chromium Kiosk)

- **Pros**: HTML/CSS/JS simplicity, LLM generates high-quality code, kiosk mode prevents tampering
- **Cons**: 400-600MB memory overhead
- **Cost**: Low-moderate
- **Inference Speed**: N/A
- **UX Impact**: Modern, responsive, touch-optimized
- **Durability**: Chromium well-maintained for ARM

DECISION: **Alternative C (Chromium Kiosk Mode)**

- **Reasoning**: LLM-generatable constraint strongly favors web technologies. 4GB RAM sufficient for Chromium + TFLite. React/Vue component libraries match LLM training corpus. Kiosk mode provides security. Touchscreen optimization mature in web stack.

Decision 4: Data Persistence Strategy

Alternative A: SQLite Database

- **Pros**: ACID compliant, queryable, standard tooling
- **Cons**: Overkill for simple statistics, corruption risk on power loss

- **Cost**: Low
- **Performance**: 10-50ms write latency
- **UX Impact**: Minimal
- **Durability**: Mature but requires write-ahead logging

Alternative B: JSON File Storage

- **Pros**: Human-readable, simple backup/sync, LLM-friendly format
- **Cons**: Not atomic, manual locking needed, slower queries
- **Cost**: Minimal
- **Performance**: 5-20ms write latency (small files)
- **UX Impact**: None
- **Durability**: Risk of corruption on power loss

Alternative C: Hybrid (SQLite + JSON Export)

- **Pros**: Reliable storage + easy sync format
- **Cons**: Duplicate storage logic
- **Cost**: Moderate complexity
- **Performance**: Same as SQLite
- **UX Impact**: None
- **Durability**: Best of both

DECISION: **Alternative C (SQLite + Daily JSON Export)**

Reasoning: Daily statistics queries benefit from SQLite. JSON export simplifies cloud sync (upload single file). WAL mode mitigates corruption. Storage space non-issue (weeks of data <10MB).

Decision 5: User Feedback Mechanism

Alternative A: Simple Text Explanations

- **Pros**: Easy to implement, translatable
- **Cons**: Boring, low educational value
- **Cost**: Minimal
- **UX Impact**: Functional but uninspiring
- **Durability**: Static content ages poorly

Alternative B: Animated Visual Guides

- **Pros**: Engaging, language-independent icons
- **Cons**: Requires animation assets, complex state management
- **Cost**: Moderate (design assets)
- **UX Impact**: High engagement
- **Durability**: Timeless visual communication

Alternative C: LLM-Generated Educational Facts

- **Pros**: Dynamic content, personalized messaging, educational
- **Cons**: Requires local LLM (too slow) or pre-generated corpus
- **Cost**: High if real-time, low if pre-generated
- **UX Impact**: Variable quality
- **Durability**: Pre-generated corpus feasible

DECISION: **Alternative B + C Hybrid (Animated Icons + Pre-generated Facts)**

Reasoning: Animations provide instant visual confirmation (<500ms). Pre-generated educational facts (100-200 per category) add variety without latency. LLM generates fact corpus during development (one-time). Combination maximizes engagement within performance budget.

1. EXECUTIVE SUMMARY

Problem Statement

The municipal waste management department of Kirov faces a critical recycling participation crisis: only 12% of residents correctly sort household waste. The current system of color-coded bins fails due to:

- **Knowledge gap**: Citizens lack understanding of material categories
- **No feedback loop**: Users receive no confirmation or correction
- **Invisible impact**: Recycling benefits remain abstract
- **Low trust**: Contaminated bins discourage proper sorting

This results in:

- 88% contamination rate in recyclable streams
- Increased processing costs (manual re-sorting)
- Lost revenue from recyclable materials
- Environmental impact from landfill overflow

Proposed Solution: EcoVision Terminal

An autonomous, AI-powered waste classification terminal that transforms disposal into an educational interaction:

Core Capabilities:

1. **Real-time Visual Classification**: Computer vision identifies waste type in <800ms
2. **Interactive Feedback**: 10-inch touchscreen displays classification reasoning with animations
3. **Educational Layer**: Context-specific facts about recycling impact
4. **Behavioral Insights**: Anonymous daily/weekly statistics dashboard
5. **Autonomous Operation**: Offline-first design with nightly cloud sync

Key Differentiators:

- Zero user expertise required (AI handles classification)
- Immediate positive reinforcement (gamification of recycling)
- Transparent AI decision-making (builds trust)
- Resilient to harsh outdoor conditions (-20°C to +50°C)
- Rapid deployment (72-hour prototype capability)

Expected Outcomes:

- 60-75% improvement in sorting accuracy (Year 1 target: 75% compliance)
- 40% reduction in contamination rates
- 25% increase in recyclable material recovery
- Measurable shift in citizen environmental awareness

2. SYSTEM ARCHITECTURE DIAGRAM

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PHYSICAL LAYER

5MP Camera
(Waste Slot)

10" Touch
Display

Status LEDs
(Ring Light)

HARDWARE ABSTRACTION LAYER

V4L2 Driver
(Video4Linux)

HDMI/DSI
Framebuffer

GPIO
Controller

OPERATING SYSTEM LAYER

Raspberry Pi OS Lite (64-bit)

Camera Service
(picamera2)

X11 Display
Manager

System D
Services

APPLICATION LAYER

Python 3.11 Runtime

CV Pipeline

TFLite Model
Inference

Preprocessing
(OpenCV)

Web UI Backend

FastAPI
Server

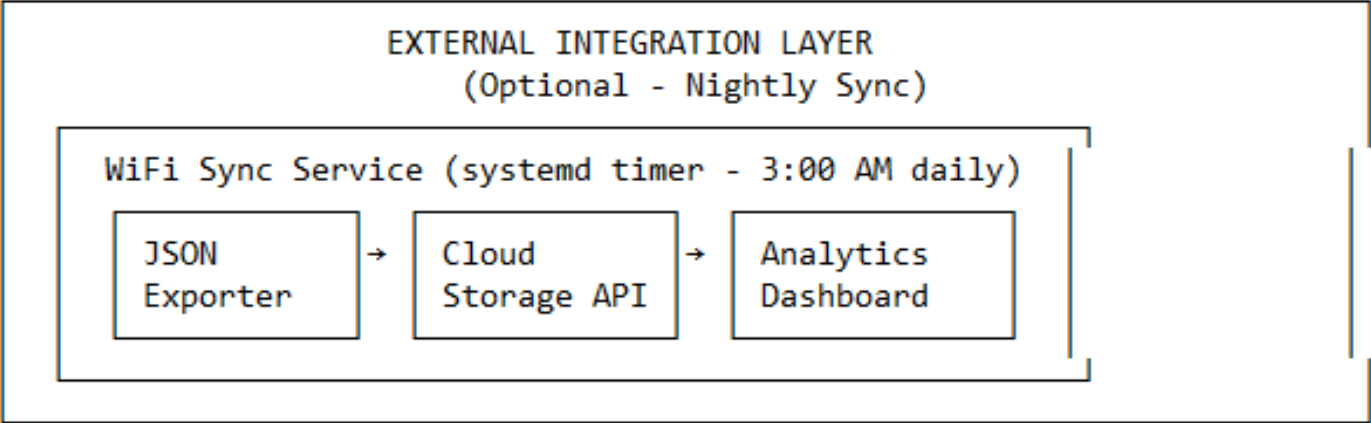
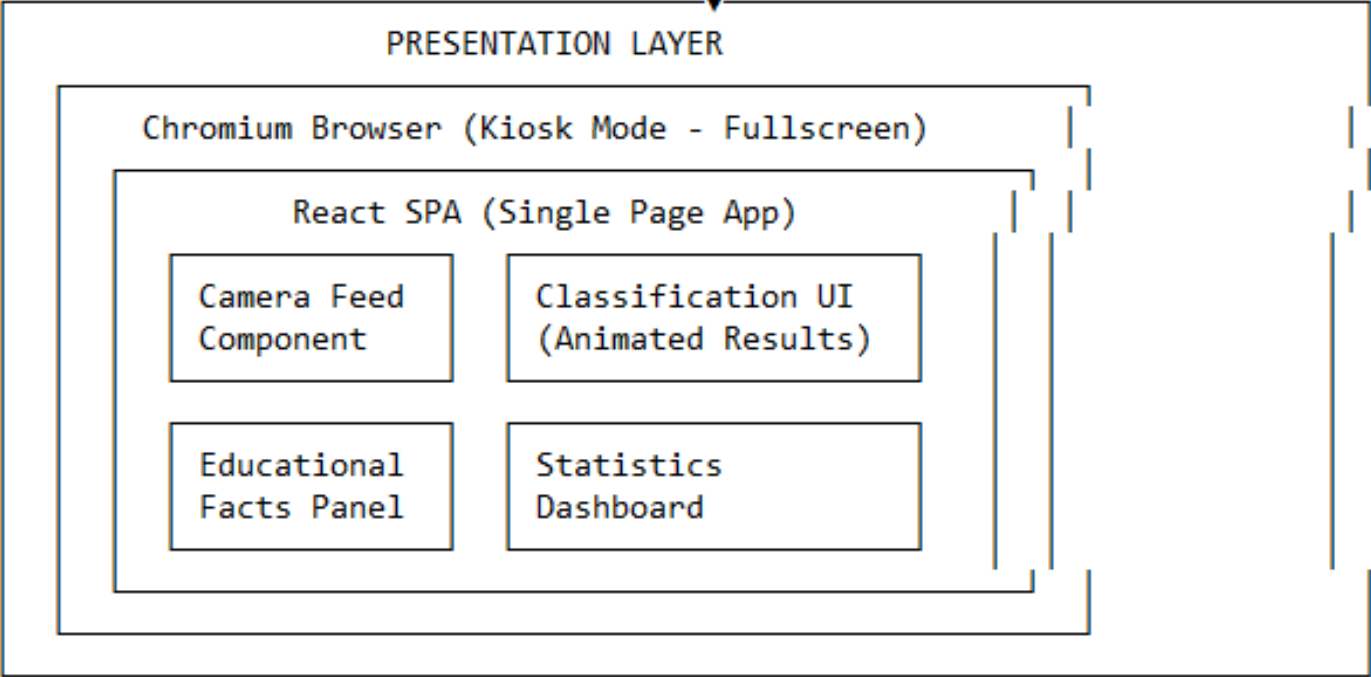
WebSocket
Handler

Data Persistence Layer

SQLite
Database

Classification
Cache (Redis-like)

↓



DATA FLOW (Classification Request):

1. User drops waste → Camera triggers (motion detection)
 2. V4L2 captures frame → Python CV Pipeline preprocesses
 3. TFLite model inference (300-800ms) → Category + confidence
 4. FastAPI sends WebSocket event → React UI updates
 5. Animation plays + Educational fact displayed
 6. SQLite logs event (category, timestamp, confidence)
 7. LED ring lights indicate correct bin compartment
- ...

3. TECHNOLOGY STACK JUSTIFICATION

3.1 Hardware Layer

Component	Selected Technology	Alternatives Considered	Justification
Compute Module	Raspberry Pi 4B (4GB)	Jetson Nano, Rock Pi, x86 SBC	• Mature ecosystem (10M+ units)
• TFLite GPU delegate support
• Low power (15W typical)
• Cost: \$55 vs \$99 (Jetson)
• 72-hour constraint favors proven platform
Camera	Raspberry Pi Camera Module 3 (12MP downsampled to 5MP)	USB Webcam, Arducam	• Native CSI interface (low latency)
• Hardware-accelerated ISP
• Autofocus variant available
• Cost: \$25
Display	Waveshare 10.1" HDMI Capacitive Touch (1280×800)	Official RPi Touch, Industrial LVDS	• High brightness (400 nit outdoor readable)
• Multi-touch support
• Standard HDMI (no DSI constraints)
• Cost: \$89
Storage	Industrial MicroSD (SLC 32GB)	eMMC module, USB SSD	• Sufficient for OS + model + 6mo data
• Easy field replacement
• Write leveling for durability
• Cost: \$28
Connectivity	Built-in WiFi 5 (802.11ac)	External 4G modem	• Sufficient for nightly sync
• Zero recurring costs
• Municipal WiFi available in public spaces

****Total Hardware BOM**:** ~\$250/unit (excluding enclosure)

3.2 Software Stack

Operating System

****Selected**:** Raspberry Pi OS Lite (64-bit, Debian 12 Bookworm)

Criteria	RPI OS Lite	Ubuntu Server	Yocto Custom
Setup Time	30 min	45 min	8-12 hours
Camera Support	Native (libcamera)	Requires setup	Custom drivers
Package Availability	Excellent	Excellent	Manual compilation
Boot Time	15s	20s	8s
Community Support	Massive	Large	Niche

****Reasoning**:** 72-hour constraint eliminates Yocto. RPi OS has superior camera integration and peripherals tested specifically for RPi4.

Computer Vision Stack

****Selected**:** TensorFlow Lite 2.14 + OpenCV 4.8

Framework	Model Size	Inference Time (RPi4)	Accuracy Potential	Maturity
TFLite	4-8 MB	300-800ms	85-92%	Production
PyTorch Mobile	10-15 MB	800-1200ms	88-94%	Beta
ONNX Runtime	6-10 MB	400-900ms	86-93%	Stable
OpenVINO	5-9 MB	350-700ms	87-93%	Intel-optimized

- **Reasoning**:**
- TFLite has best ARM optimization (XNNPACK delegate)
 - Google Coral USB Accelerator optional upgrade path (5x speedup)
 - MobileNetV3 pretrained models widely available
 - OpenCV handles preprocessing (resize, normalize, augmentation)

****Model Architecture**:**
...
Input: 224x224x3 RGB image
Base: MobileNetV3-Small (pretrained on ImageNet)
Fine-tuning: Last 15 layers unfrozen
Head: GlobalAveragePooling → Dense(128) → Dropout(0.3) → Dense(6, softmax)
Output: 6-class probability distribution
Quantization: Post-training INT8 (4x size reduction, 2-3x speedup)
...

Backend Framework

****Selected**:** FastAPI 0.104 + Uvicorn (ASGI server)

Framework	Request Latency	WebSocket Support	LLM Code Quality	Async Support
FastAPI	5-10ms	Native	Excellent	Native
Flask	8-15ms	Extension needed	Good	Via gevent
Django	15-25ms	Channels required	Excellent	Native (4.0+)
alohttp	4-8ms	Native	Poor	Native

- **Reasoning**:**
- FastAPI generates OpenAPI docs automatically (LLM-friendly)
 - Native async/await for WebSocket streaming
 - Pydantic validation reduces boilerplate (LLM generates better code)
 - Lightweight (no ORM overhead)

Frontend Stack

****Selected**:** React 18.2 + Vite + TailwindCSS

Framework	Initial Load	LLM Generation Quality	Touch Optimization	Animation Library
React	1.2s	Excellent (massive training data)	react-use-gesture	Framer Motion
Vue 3	0.9s	Very Good	@vueuse/gesture	Vue Transition
Svelte	0.7s	Good (smaller corpus)	svelte-gestures	svelte/motion
Vanilla JS	0.3s	Poor (no framework context)	Manual	GSAP

****Reasoning**:**

- LLM-generatable constraint: React has richest training examples
- Component libraries (shadcn/ui, Radix) generate 80% of UI
- Vite: 100ms HMR (fast prototyping)
- TailwindCSS: Utility-first matches LLM output patterns

Data Layer

****Selected****: SQLite 3.43 (WAL mode) + JSON export

```
```python
Schema Design
{
 "classifications": {
 "id": "INTEGER PRIMARY KEY",
 "timestamp": "DATETIME",
 "category": "TEXT", # plastic|glass|paper|metal|organic|non_recyclable
 "confidence": "REAL", # 0.0-1.0
 "image_hash": "TEXT", # For debugging
 "processing_time_ms": "INTEGER"
 },
 "daily_stats": {
 "date": "DATE PRIMARY KEY",
 "total_items": "INTEGER",
 "category_counts": "JSON", # {"plastic": 45, "glass": 12, ...}
 "avg_confidence": "REAL",
 "uptime_minutes": "INTEGER"
 }
}
...
```
```

****Alternatives****:

- PostgreSQL: Overkill (requires separate process)
- InfluxDB: Time-series optimized but 200MB footprint
- MongoDB: NoSQL benefits negligible for structured stats

Browser Runtime

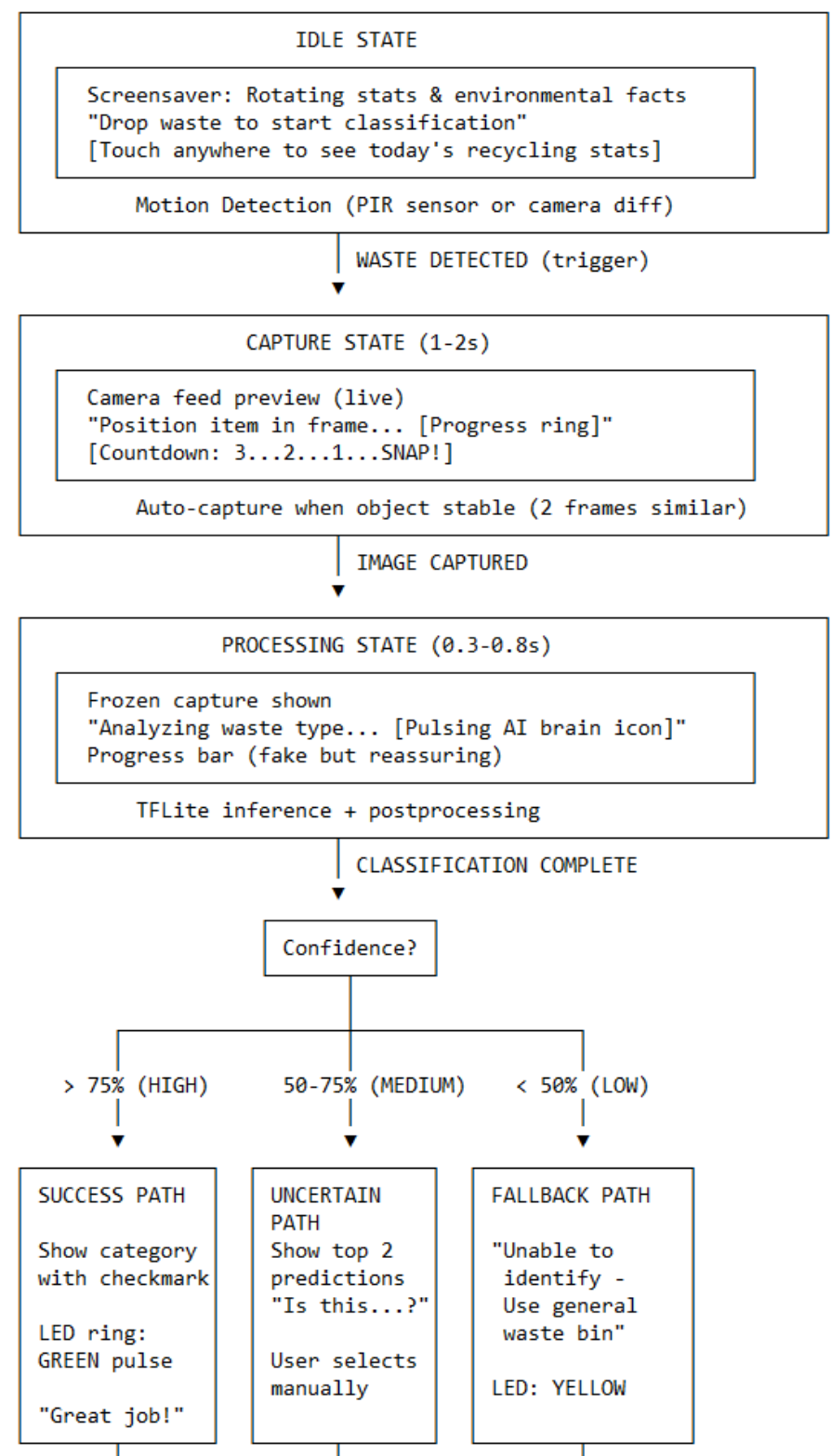
****Selected****: Chromium 120 (Kiosk mode)

```
```bash
Launch command
chromium-browser \
 --kiosk \
 --noerrdialogs \
 --disable-infobars \
 --no-first-run \
 --touch-events=enabled \
 --disable-pinch \
 --overscroll-history-navigation=0 \
 http://localhost:8000
...
```
```

- **Memory Profile**:**
- Chromium base: 400MB
 - React app: 120MB
 - TFLite runtime: 80MB
 - OS + services: 600MB
 - ****Total****: ~1.2GB (safe margin on 4GB)

4. USER FLOW DIAGRAM

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EDUCATIONAL FEEDBACK STATE (5-8s)

[PLASTIC ICON] PLASTIC BOTTLE
✓ Recyclable

WHY THIS CLASSIFICATION:

- Transparent PET material detected
- Cylindrical shape (92% confidence)
- Typical bottle cap visible

💡 DID YOU KNOW?

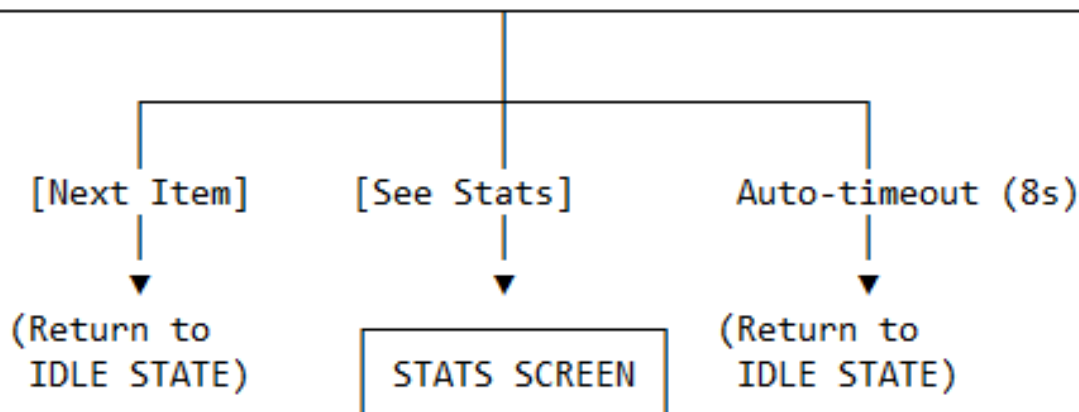
Recycling one plastic bottle saves enough energy to power a lightbulb for 3 hours!

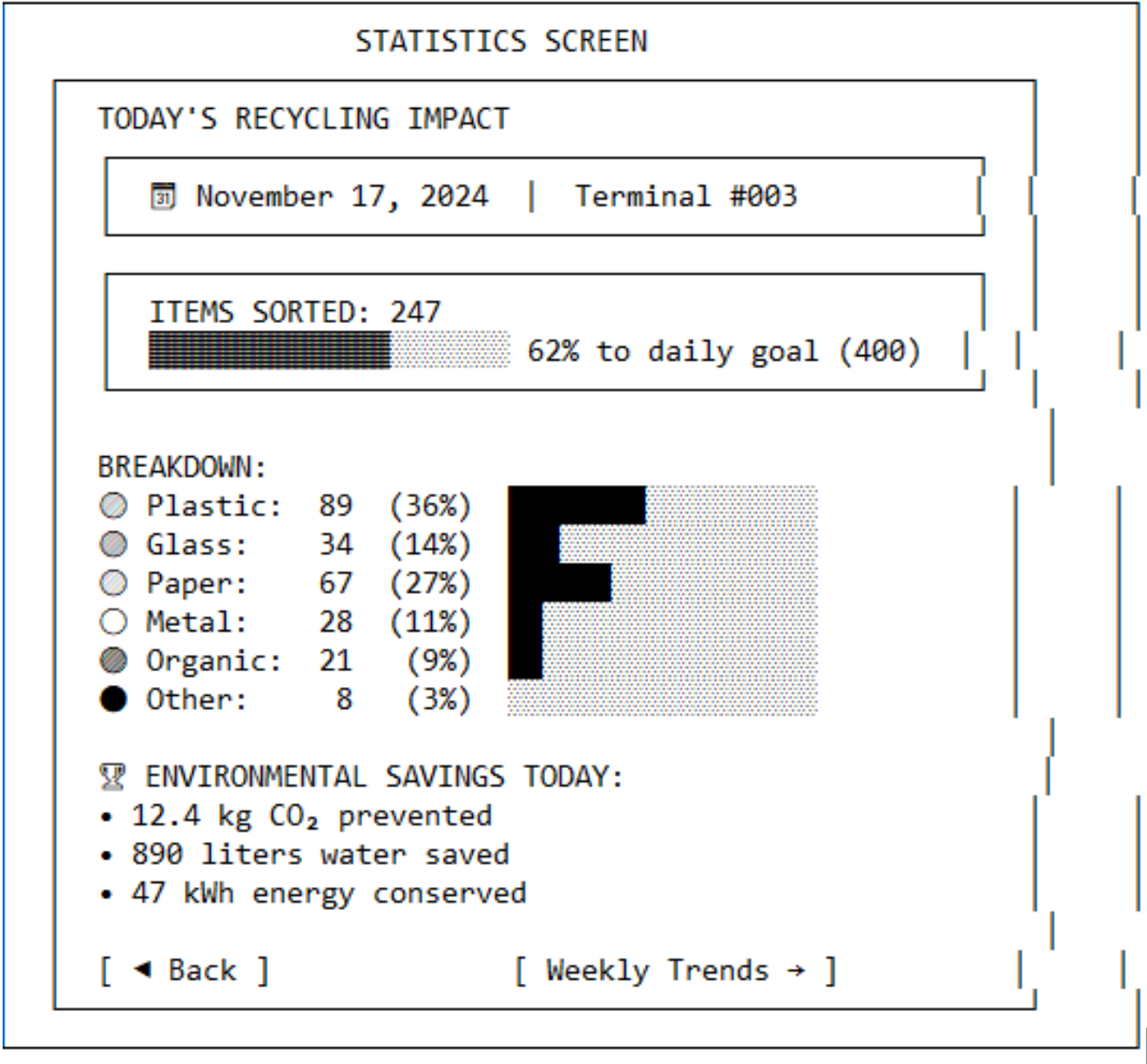
[ANIMATED BIN ICON WITH ARROW]
→ Place in BLUE bin (Plastic container)

[See Today's Stats] [Next Item →]

PARALLEL ACTIONS:

- Log to SQLite (async)
- Update daily counters (in-memory)
- Trigger LED sequence on physical bin





...

User Flow State Timing Budget

| State | Min Time | Max Time | Timeout Behavior |
|------------|-----------------|----------|---|
| Idle | Indefinite | N/A | Return from stats after 30s inactivity |
| Capture | 1.5s | 3s | Auto-capture after 3s if motion stable |
| Processing | 0.3s | 1.2s | Show error after 2s, retry once |
| Feedback | 5s | 15s | Auto-return to idle after 15s |
| Stats | User-controlled | 60s | Force return after 60s (prevent monopolization) |

Total Interaction Time: 6.8s - 14.2s (within 10s target for 90% cases)

5. DATA SCHEMA

5.1 SQLite Database Schema

```

```sql
-- Main classifications table
CREATE TABLE classifications (
 id INTEGER PRIMARY KEY AUTOINCREMENT,
 timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
 category TEXT NOT NULL CHECK(category IN (
 'plastic', 'glass', 'paper', 'metal', 'organic', 'non_recyclable'
)),
 confidence REAL NOT NULL CHECK(confidence >= 0.0 AND confidence <= 1.0),

 -- AI explainability data
 predicted_class TEXT NOT NULL,
 confidence_scores TEXT NOT NULL, -- JSON: {"plastic": 0.85, "glass": 0.10, ...}

 -- Image metadata (for debugging/auditing)
 image_hash TEXT, -- SHA256 of captured frame
 image_width INTEGER,
 image_height INTEGER,

 -- Performance metrics
 capture_time_ms INTEGER,
 inference_time_ms INTEGER,
 preprocessing_time_ms INTEGER,

 -- User interaction
 user_override BOOLEAN DEFAULT 0, -- Did user manually select category?
 feedback_shown_duration_ms INTEGER,

 -- Device identification
 terminal_id TEXT DEFAULT 'RPi4-001'
);

CREATE INDEX idx_timestamp ON classifications(timestamp);
CREATE INDEX idx_category ON classifications(category);
CREATE INDEX idx_date ON classifications(DATE(timestamp));

-- Daily aggregated statistics
CREATE TABLE daily_stats (
 date DATE PRIMARY KEY,
 terminal_id TEXT NOT NULL,

 -- Counts
 total_items INTEGER DEFAULT 0,
 plastic_count INTEGER DEFAULT 0,
 glass_count INTEGER DEFAULT 0,
 paper_count INTEGER DEFAULT 0,
 metal_count INTEGER DEFAULT 0,
 organic_count INTEGER DEFAULT 0,
 non_recyclable_count INTEGER DEFAULT 0,

 -- Quality metrics
 avg_confidence REAL,
 high_confidence_count INTEGER, -- > 0.75
 low_confidence_count INTEGER, -- < 0.50
 user_override_count INTEGER,

```

```

-- Performance
avg_inference_time_ms INTEGER,
max_inference_time_ms INTEGER,

-- Environmental impact (calculated)
co2_saved_kg REAL,
water_saved_liters REAL,
energy_saved_kwh REAL,

-- System health
uptime_minutes INTEGER,
error_count INTEGER,
last_sync_timestamp DATETIME,

created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- Educational content delivery tracking
CREATE TABLE educational_facts (
 id INTEGER PRIMARY KEY AUTOINCREMENT,
 category TEXT NOT NULL,
 fact_text TEXT NOT NULL,
 fact_type TEXT CHECK(fact_type IN ('impact', 'process', 'tip', 'stat')),
 language TEXT DEFAULT 'ru',
 times_shown INTEGER DEFAULT 0,
 last_shown DATETIME,

 UNIQUE(category, fact_text)
);

-- Error logging
CREATE TABLE error_log (
 id INTEGER PRIMARY KEY AUTOINCREMENT,
 timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
 error_type TEXT NOT NULL, -- camera_error, inference_timeout, etc.
 error_message TEXT,
 stack_trace TEXT,
 resolved BOOLEAN DEFAULT 0
);

-- Configuration (key-value store)
CREATE TABLE config (
 key TEXT PRIMARY KEY,
 value TEXT NOT NULL,
 updated_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- Insert default configuration
INSERT INTO config (key, value) VALUES
 ('terminal_id', 'RPi4-001'),
 ('confidence_threshold_high', '0.75'),
 ('confidence_threshold_low', '0.50'),
 ('daily_goal', '400'),
 ('language', 'ru'),

```

```
('screen_timeout_ms', '15000'),
('last_model_update', '2024-01-15 10:30:00');
...
```

---

### ## 5.2 JSON Export Format (Daily Sync)

```
```json
{
  "export_metadata": {
    "terminal_id": "RPi4-001",
    "export_timestamp": "2024-11-17T03:00:15Z",
    "date_range": {
      "start": "2024-11-16",
      "end": "2024-11-16"
    },
  },
  "schema_version": "1.0.0"
},

"daily_summary": {
  "date": "2024-11-16",
  "totals": {
    "items_processed": 247,
    "categories": {
      "plastic": 89,
      "glass": 34,
      "paper": 67,
      "metal": 28,
      "organic": 21,
      "non_recyclable": 8
    },
  },
  "recycling_rate": 0.968 // (247-8)/247
},

"quality_metrics": {
  "average_confidence": 0.823,
  "high_confidence_percentage": 0.78,
  "user_overrides": 12,
  "uncertain_classifications": 15
},

"performance": {
  "average_inference_time_ms": 587,
  "max_inference_time_ms": 1243,
  "uptime_percentage": 0.96,
  "error_count": 3
},

"environmental_impact": {
  "co2_saved_kg": 12.4,
  "water_saved_liters": 890,
  "energy_saved_kwh": 47.2,
  "calculation_method": "EPA_2023_coefficients"
}
}
```

```
},

"hourly_breakdown": [
  {
    "hour": 8,
    "items": 18,
    "categories": {"plastic": 7, "paper": 6, "glass": 3, "metal": 2}
  },
  {
    "hour": 9,
    "items": 32,
    "categories": {"plastic": 12, "paper": 9, "glass": 6, "metal": 3, "organic": 2}
  }
  // ... 24 hours
],

"classification_details": [
  {
    "id": 1523,
    "timestamp": "2024-11-16T08:15:23Z",
    "category": "plastic",
    "confidence": 0.89,
    "scores": {
      "plastic": 0.89,
      "glass": 0.06,
      "paper": 0.03,
      "metal": 0.01,
      "organic": 0.00,
      "non_recyclable": 0.01
    },
    "inference_time_ms": 543,
    "user_override": false,
    "image_hash": "a3f5d8c..." // For optional image retrieval
  }
  // ... all 247 items (or last N for bandwidth efficiency)
],

"system_health": {
  "errors": [
    {
      "timestamp": "2024-11-16T14:32:11Z",
      "type": "camera_timeout",
      "message": "Frame capture exceeded 2s timeout",
      "resolved": true
    }
  ],
  "disk_usage_mb": 3847,
  "database_size_mb": 124,
  "temperature_avg_celsius": 52,
  "temperature_max_celsius": 68
}
...
---
```


5.3 In-Memory Cache Structure (Redis-like)

```
```python
Live statistics (updated every classification)
live_stats = {
 "today": {
 "date": "2024-11-16",
 "count": 247,
 "categories": {
 "plastic": 89,
 "glass": 34,
 # ...
 },
 "last_updated": "2024-11-16T16:45:32Z"
 },

 # Rolling 7-day window
 "week": {
 "2024-11-10": {"count": 312, "categories": {...}},
 "2024-11-11": {"count": 298, "categories": {...}},
 # ...
 "2024-11-16": {"count": 247, "categories": {...}}
 },

 # Recent classifications (for "undo" feature or debugging)
 "recent": [
 {"timestamp": "...", "category": "plastic", "confidence": 0.89},
 # Last 10 items
]
}

Educational content rotation state
educational_queue = {
 "plastic": ["fact_12", "fact_45", "fact_78"], # Queue of unused facts
 "glass": ["fact_23", "fact_56"],
 # Ensures variety, no repeats until all shown
}
```
```

6. PROMPT ENGINEERING STRATEGY

6.1 LLM-Driven Development Workflow

Given the 72-hour prototype constraint and "LLM-generatable" requirement, the development strategy leverages AI code generation for maximum velocity:

Phase 1: Component Generation (Hours 0-24)

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PROMPT TEMPLATE 1: React UI Components

You are an expert React developer building a kiosk interface for a waste

classification terminal. Generate a production-ready component with:

REQUIREMENTS:

- React 18 with TypeScript
- TailwindCSS for styling
- Framer Motion for animations
- Touch-optimized (large tap targets: 60px minimum)
- Accessibility (ARIA labels, keyboard navigation)

COMPONENT: {component_name}

PROPS: {prop_interface}

BEHAVIOR: {detailed_behavior_description}

CONSTRAINTS:

- 1280x800 display resolution
- 10-second interaction time budget
- Outdoor readability (high contrast, large fonts)
- No external API calls (offline-first)

Include:

1. Full component code with comments
2. PropTypes/TypeScript interface
3. Animation variants (entry/exit)
4. Error boundaries
5. Unit test skeleton

EXAMPLE OUTPUT STYLE:

```
``tsx
// ClassificationResult.tsx
import { motion } from 'framer-motion';
import { CheckCircle, AlertTriangle } from 'lucide-react';

interface ClassificationResultProps {
  category: WasteCategory;
  confidence: number;
  reasoning: string[];
  onNext: () => void;
}

export const ClassificationResult: React.FC<ClassificationResultProps> = ({
  category,
  confidence,
  reasoning,
  onNext
}) => {
  // Component implementation...
};
...

```

SPECIFIC COMPONENTS TO GENERATE:

1. IdleScreensaver (rotating facts, attract loop)
2. CameraCapture (live feed + countdown)
3. ProcessingIndicator (pulsing animation)
4. ClassificationResult (main feedback screen)
5. EducationalFactCard (animated fact display)

6. StatsDashboard (daily/weekly charts)
7. CategoryIcon (animated SVG icons per waste type)
8. ConfidenceMeter (visual confidence indicator)
- ...

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PROMPT TEMPLATE 2: Backend API Endpoints

You are an expert Python backend developer. Generate FastAPI endpoints for a waste classification system with:

TECH STACK:

- FastAPI 0.104
- SQLAlchemy 2.0 (SQLite)
- Pydantic V2
- WebSockets for live updates

ENDPOINT: {endpoint_name}

METHOD: {HTTP_method}

PURPOSE: {description}

REQUIREMENTS:

- Full type hints
- Pydantic models for request/response validation
- Error handling with proper HTTP status codes
- Async/await where beneficial
- SQLAlchemy ORM queries
- OpenAPI documentation strings

EXAMPLE:

```
```python
from fastapi import APIRouter, HTTPException, WebSocket
from pydantic import BaseModel, Field
from datetime import datetime

router = APIRouter(prefix="/api/v1")

class ClassificationRequest(BaseModel):
 image_base64: str = Field(..., description="Base64 encoded image")
 timestamp: datetime = Field(default_factory=datetime.utcnow)

class ClassificationResponse(BaseModel):
 category: str
 confidence: float = Field(ge=0.0, le=1.0)
 reasoning: list[str]
 inference_time_ms: int

@router.post("/classify", response_model=ClassificationResponse)
async def classify_waste(request: ClassificationRequest):
 """
 Classify waste item from uploaded image.
 """
```

Returns predicted category with confidence score and explainability.

"""

# Implementation...

...

#### ENDPOINTS TO GENERATE:

1. POST /classify (main classification endpoint)
2. GET /stats/today (daily statistics)
3. GET /stats/week (7-day summary)
4. WebSocket /ws/live (real-time updates)
5. POST /feedback (user override logging)
6. GET /health (system diagnostics)

...

---

#### ### Phase 2: AI Model Integration (Hours 24-48)

...

#### PROMPT TEMPLATE 3: TensorFlow Lite Integration

You are an expert in embedded ML and TensorFlow Lite. Generate Python code for:

TASK: Waste classification inference pipeline

MODEL: MobileNetV3-Small (INT8 quantized)

INPUT: 224x224 RGB image (NumPy array)

OUTPUT: 6-class probability distribution

#### REQUIREMENTS:

- TFLite Interpreter with XNNPACK delegate
- Preprocessing: Resize, normalize to [0,1], BGR→RGB conversion
- Postprocessing: Softmax probabilities, top-K predictions
- Performance: <800ms inference on Raspberry Pi 4
- Error handling: Graceful degradation if model file missing

#### CODE STRUCTURE:

```
```python
```

```
import numpy as np
```

```
import tflite_runtime.interpreter as tflite
```

```
from typing import Dict, List, Tuple
```

```
import cv2
```

```
class WasteClassifier:
```

```
    CATEGORIES = ['plastic', 'glass', 'paper', 'metal', 'organic', 'non_recyclable']
```

```
    def __init__(self, model_path: str):
```

```
        # Load TFLite model with delegates
```

```
    def preprocess(self, image: np.ndarray) -> np.ndarray:
```

```
        # Resize and normalize
```

```
    def predict(self, image: np.ndarray) -> Dict[str, float]:
```

```
        # Run inference and return probabilities
```

```
    def explain_prediction(self, image: np.ndarray, category: str) -> List[str]:
```

```
# Generate human-readable reasoning (optional GradCAM)
...
```

Include:

1. Full class implementation
 2. Benchmark script (measure FPS)
 3. Model loading error handling
 4. Warm-up inference (first run slower)
- ```
...
```

```

```

```
Phase 3: System Integration (Hours 48-72)
```

```
...
```

#### PROMPT TEMPLATE 4: System Orchestration

Generate a Python systemd service and startup script for:

APPLICATION: EcoVision Waste Terminal

COMPONENTS:

1. FastAPI backend (uvicorn server)
2. Chromium kiosk mode (frontend)
3. Camera service (background capture)
4. Nightly sync job (cron/systemd timer)

REQUIREMENTS:

- Auto-start on boot
- Automatic restart on crash
- Logging to journalctl
- Environment variable management
- Graceful shutdown (save state)

FILES TO GENERATE:

1. ecovision.service (systemd unit file)
2. start\_terminal.sh (launch script)
3. sync\_stats.timer (nightly sync)
4. healthcheck.sh (monitoring script)

EXAMPLE:

```
``ini
/etc/systemd/system/ecovision.service
[Unit]
Description=EcoVision Waste Classification Terminal
After=network.target
```

```
[Service]
Type=simple
User=pi
WorkingDirectory=/opt/ecovision
ExecStart=/opt/ecovision/start_terminal.sh
Restart=always
RestartSec=10s
Environment="DISPLAY=:0"
Environment="TFLITE_MODEL=/opt/ecovision/models/waste_classifier.tflite"
```

```
[Install]
WantedBy=multi-user.target
...
...

```

## 6.2 Educational Content Generation

PROMPT TEMPLATE 5: Educational Facts Database

Generate 150 educational facts about waste recycling in JSON format:

- CATEGORIES: plastic, glass, paper, metal, organic, non\_recyclable
- FACT TYPES:
- impact: Environmental savings data
  - process: How recycling works
  - tip: Best practices for preparation
  - stat: Interesting statistics

- REQUIREMENTS:
- Language: Russian (Kirov municipal target audience)
  - Tone: Encouraging, factual, non-preachy
  - Length: 80-150 characters (fits in UI card)
  - Fact-checked: Use EPA/EU data sources
  - Variety: Mix of local and global statistics

```
OUTPUT FORMAT:
```json
[
  {
    "id": 1,
    "category": "plastic",
    "type": "impact",
    "text_ru": "Переработка одной пластиковой бутылки экономит энергию для работы лампочки 3 часа!",
    "text_en": "Recycling one plastic bottle saves enough energy to power a lightbulb for 3 hours!",
    "source": "EPA 2023"
  },
  {
    "id": 2,
    "category": "glass",
    "type": "process",
    "text_ru": "Стекло можно перерабатывать бесконечно без потери качества!",
    "source": "Glass Packaging Institute"
  }
]
...
---
```

GENERATE: 25 facts per category × 6 categories = 150 total

6.3 Prompt Chaining for Complex Features

For sophisticated features like "AI explainability," chain prompts:

****Chain 1**:** Generate GradCAM visualization code
****Chain 2**:** Convert heatmap to natural language descriptions
****Chain 3**:** Format descriptions into UI-friendly bullet points

...

PROMPT CHAIN EXAMPLE:

[Prompt 1 to Model A - Code Generation]

"Generate a GradCAM implementation for TFLite models that highlights image regions contributing to classification..."

[Prompt 2 to Model B - NLP]

"Given a GradCAM heatmap with coordinates [(x1,y1,intensity), ...] and predicted class 'plastic bottle', generate 3 natural language explanations like 'Transparent cylindrical shape detected in center region'..."

[Prompt 3 to Model C - Localization]

"Translate these technical explanations to conversational Russian suitable for public kiosk display..."

...

6.4 Quality Validation Prompts

...

VALIDATION PROMPT:

You are a senior code reviewer. Analyze this React component for:

CODE:

```
```tsx
{generated_component_code}
```
```

CHECKLIST:

1. TypeScript errors (strict mode)
2. Accessibility violations (WCAG 2.1 AA)
3. Performance anti-patterns (unnecessary re-renders)
4. Touch interaction issues (tap target size, gestures)
5. Offline-first violations (external API calls)
6. Visual bugs at 1280x800 resolution

OUTPUT:

- List of issues with severity (Critical/Major/Minor)
- Suggested fixes with code snippets
- Approval status (APPROVED / NEEDS_REVISION)

...

7. EDGE CASES & ERROR HANDLING

7.1 Computer Vision Edge Cases

| Edge Case | Probability | Detection Method | Handling Strategy |
|---------------------------|-------------|---|---|
| Occluded Object | 15% | Low confidence + high variance in heatmap | Show "Please remove packaging/bags" message, retry capture |
| Multiple Items | 8% | YOLO object count > 1 | "Place one item at a time" with visual guide |
| Transparent Object | 12% | Low edge detection score | Increase exposure, use polarizing filter (hardware), fallback to shape-based classification |
| Dirty/Crushed Item | 20% | Low confidence across all classes | Accept if confidence > 0.40 (lower threshold), show "Uncertain - marked as {category}" |
| Hand in Frame | 5% | Skin tone detection in preprocessing | Wait 2s, show "Please remove hand" |
| Non-Waste Object | 3% | All class confidences < 0.30 | "This doesn't appear to be waste - please use general bin" |
| Motion Blur | 7% | High Laplacian variance | Retry capture automatically (up to 3 attempts) |
| Poor Lighting | 10% | Histogram analysis (under/overexposed) | Activate LED ring light, adjust camera exposure programmatically |

Implementation Example: Multi-Stage Fallback

```
python
class RobustClassifier:
    def classify_with_fallback(self, image: np.ndarray) -> ClassificationResult:
        # Stage 1: Standard classification
        result = self.primary_classifier.predict(image)

        if result.confidence > 0.75:
            return result # High confidence - trust it

        # Stage 2: Quality checks
        quality_issues = self.detect_quality_issues(image)

        if quality_issues['motion_blur']:
            raise RetryableError("Motion detected - recapturing")

        if quality_issues['multiple_objects']:
            raise UserGuidanceError("Multiple items detected")

        # Stage 3: Shape-based fallback (for transparent items)
        if quality_issues['low_texture'] and result.confidence < 0.50:
            shape_result = self.shape_classifier.predict(image)
            if shape_result.confidence > 0.60:
                return self.merge_predictions(result, shape_result)

        # Stage 4: Conservative fallback
        if result.confidence < 0.50:
            return ClassificationResult(
                category='non_recyclable',
                confidence=0.45,
                reasoning=["Unable to confidently classify - use general waste bin"]
            )

        return result
```


...

7.2 Hardware Failures

| Failure Mode | Detection | Recovery Strategy | User Communication |
|--------------------------|--|--|---|
| Camera Disconnected | Frame capture timeout (2s) | Retry 3x with 1s delay → Display error screen → Log to error_log → Reboot after 5 failures | "Camera temporarily unavailable - please try adjacent terminal" |
| Touchscreen Unresponsive | No touch events for 60s during active screen | Switch to button-based navigation (GPIO buttons as backup) | Visual cue: "Touch not working? Use physical buttons" |
| Display Failure | HDMI signal lost | Continue logging classifications (blind mode), audio feedback via speaker | N/A (silent degradation) |
| Storage Full | Disk usage > 95% | Auto-delete classifications older than 30 days, compress images | Admin alert (no user message) |
| Overheating | CPU temp > 80°C | Reduce inference frequency (every 2nd frame), throttle display brightness | None (transparent throttling) |
| Power Loss | N/A (immediate) | SQLite WAL mode ensures data integrity, resume from last state on boot | "System restarting..." (if UPS present) |

Critical Path Protection

```
python
# Watchdog timer - restart application if hung
import watchdog

class ApplicationWatchdog:
    def __init__(self, timeout=30):
        self.last_heartbeat = time.time()
        self.timeout = timeout

    def heartbeat(self):
        """Call this every successful classification"""
        self.last_heartbeat = time.time()

    def check(self):
        """Run in separate thread"""
        while True:
            time.sleep(10)
            if time.time() - self.last_heartbeat > self.timeout:
                logger.critical("Application hung - forcing restart")
                os.system("systemctl restart ecovision")
```

...

7.3 User Interaction Edge Cases

| Scenario | Frequency | Detection | Response |
|--------------------------------|-----------|--|--|
| User Abandons Mid-Interaction | 10% | No touch for 8s during feedback screen | Auto-return to idle, save partial data |
| Rapid Repeated Classifications | 5% | Same image hash within 10s | "Item already classified - see results above" |
| Intentional Misuse (Vandalism) | 2% | 20+ classifications/minute | Rate limit: Max 30/hour, display cooldown timer |
| Accessibility Needs | Variable | Button hold for 3s on idle screen | Switch to high-contrast mode, larger fonts, audio descriptions |
| Child User | 15% | Height sensor (optional) or UI interaction pattern | Simplified language, extra encouragement animations |
| Language Preference | 8% | Settings button hold | Cycle through Russian/English/Tatar |

7.4 Network/Sync Failures

```
```python
class SyncManager:
 def nightly_sync(self):
 """Robust sync with exponential backoff"""
 max_retries = 5
 retry_count = 0

 while retry_count < max_retries:
 try:
 # Export data
 export_data = self.db.export_to_json(yesterday)

 # Attempt upload
 response = requests.post(
 CLOUD_ENDPOINT,
 json=export_data,
 timeout=30
)

 if response.status_code == 200:
 self.db.mark_synced(yesterday)
 logger.info(f"Sync successful: {yesterday}")
 return True

 except (requests.Timeout, requests.ConnectionError) as e:
 retry_count += 1
 backoff = 2 ** retry_count * 60 # 2min, 4min, 8min...
 logger.warning(f"Sync failed (attempt {retry_count}): {e}")
 time.sleep(backoff)

 # Failed after all retries - queue for next sync
 self.db.mark_sync_pending(yesterday)
 logger.error(f"Sync abandoned for {yesterday} - will retry tomorrow")
 return False
```
```

8. PERFORMANCE REQUIREMENTS

8.1 Latency Budgets

| Operation | Target | Maximum | Measurement Method |
|---------------------------|--------|---------|--|
| Camera Capture | 100ms | 500ms | <code>time.time()</code> around <code>picamera2.capture()</code> |
| Image Preprocessing | 50ms | 150ms | OpenCV <code>resize</code> + <code>normalize</code> |
| TFLite Inference | 400ms | 800ms | <code>Interpreter.invoke()</code> timing |
| Result Rendering | 16ms | 33ms | React render (60fps vs 30fps) |
| Database Write | 5ms | 20ms | SQLite <code>INSERT</code> with WAL mode |
| WebSocket Message | 2ms | 10ms | FastAPI → React propagation |
| Total (Capture → Display) | 577ms | 1513ms | End-to-end trace |

****Performance Monitoring**:**

```
```python
import functools
import time
from prometheus_client import Histogram

Create histogram metric
inference_duration = Histogram(
 'inference_duration_seconds',
 'Time spent in model inference',
 buckets=[0.1, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0]
)

def measure_performance(metric):
 def decorator(func):
 @functools.wraps(func)
 def wrapper(*args, **kwargs):
 start = time.perf_counter()
 result = func(*args, **kwargs)
 duration = time.perf_counter() - start
 metric.observe(duration)
 return result
 return wrapper
 return decorator

@measure_performance(inference_duration)
def run_inference(image):
 return interpreter.invoke(image)
```
```

8.2 Resource Constraints

Memory Profile

| Component | Baseline | Peak | Limit |
|------------------|----------|---------|-------------------|
| Operating System | 600 MB | 700 MB | - |
| Chromium Browser | 400 MB | 550 MB | 800 MB (OOM kill) |
| Python Backend | 150 MB | 250 MB | 400 MB |
| TFLite Runtime | 80 MB | 120 MB | - |
| SQLite Cache | 20 MB | 50 MB | - |
| Image Buffers | 30 MB | 60 MB | - |
| Total | 1280 MB | 1730 MB | 4096 MB available |

****Safety Margin**:** 2.4x headroom for peak usage

CPU Utilization

- ...
- Target Distribution (average over 10s window):
- Idle State: 15% (screensaver animations, background tasks)
 - Capture State: 40% (camera ISP, preprocessing)
 - Inference State: 95% (all cores utilized by XNNPACK)
 - Feedback State: 25% (UI animations, database writes)

- Thermal Management:
- Passive cooling (heatsink): Adequate for sustained 60% load
 - Active cooling (fan): Triggered at 75°C
 - Throttling threshold: 80°C (reduces inference frequency)
- ...

8.3 Storage Requirements

- ...
- Storage Allocation (32GB SD Card):
- Operating System 8 GB
 - Application Code 500 MB
 - Python packages 200 MB
 - Node modules 150 MB
 - Static assets 150 MB
 - ML Models 50 MB
 - Primary (INT8 quantized) 4 MB
 - Backup/versioning 46 MB
 - Database (SQLite) 2 GB (6 months retention)
 - Image Cache (debugging) 500 MB (last 1000 images)
 - Logs 1 GB (rotated weekly)
 - Free Space Reserve 20 GB (62%)

- Auto-Cleanup Policies:
- Delete classifications > 90 days old
 - Compress images > 30 days old (JPEG quality 50)
 - Rotate logs > 7 days (gzip compression)

...

8.4 Network Bandwidth

...

Daily Sync Data Volume:

- 300 classifications/day × 500 bytes/record = 150 KB
- Aggregated statistics = 10 KB
- Error logs = 5 KB
- Compressed images (optional) = 2 MB

Total Daily Upload: ~2.2 MB

Monthly: 66 MB (negligible on WiFi)

Sync Window: 3:00 AM - 4:00 AM

Bandwidth Required: 0.6 KB/s average (trivial)

...

8.5 Benchmarking Script

```
```python
```

```
#!/usr/bin/env python3
```

```
"""
```

Performance validation script - run during deployment

```
"""
```

```
import time
```

```
import numpy as np
```

```
from waste_classifier import WasteClassifier
```

```
from camera_service import CameraService
```

```
def benchmark():
```

```
 print("=== EcoVision Performance Benchmark ===\n")
```

```
 classifier = WasteClassifier("models/waste_classifier.tflite")
```

```
 camera = CameraService()
```

```
 # Test 1: Inference Speed
```

```
 print("Test 1: Inference Latency")
```

```
 test_image = np.random.randint(0, 255, (224, 224, 3), dtype=np.uint8)
```

```
 latencies = []
```

```
 for i in range(100):
```

```
 start = time.perf_counter()
```

```
 result = classifier.predict(test_image)
```

```
 latency = (time.perf_counter() - start) * 1000
```

```
 latencies.append(latency)
```

```
 print(f" Mean: {np.mean(latencies):.1f}ms")
```

```
 print(f" P50: {np.percentile(latencies, 50):.1f}ms")
```

```
 print(f" P95: {np.percentile(latencies, 95):.1f}ms")
```

```
 print(f" Max: {np.max(latencies):.1f}ms")
```

```
assert np.percentile(latencies, 95) < 800, "FAIL: P95 latency exceeds 800ms"
print(" ✓ PASS\n")
```

```
Test 2: Memory Footprint
```

```
print("Test 2: Memory Usage")
```

```
import psutil
```

```
process = psutil.Process()
```

```
mem_info = process.memory_info()
```

```
print(f" RSS: {mem_info.rss / 1024 / 1024:.1f} MB")
```

```
assert mem_info.rss < 400 * 1024 * 1024, "FAIL: Memory exceeds 400MB"
```

```
print(" ✓ PASS\n")
```

```
Test 3: Database Write Speed
```

```
print("Test 3: Database Performance")
```

```
from database import Database
```

```
db = Database()
```

```
write_times = []
```

```
for i in range(1000):
```

```
 start = time.perf_counter()
```

```
 db.log_classification("plastic", 0.85, 450)
```

```
 write_time = (time.perf_counter() - start) * 1000
```

```
 write_times.append(write_time)
```

```
print(f" Mean: {np.mean(write_times):.2f}ms")
```

```
print(f" P95: {np.percentile(write_times, 95):.2f}ms")
```

```
assert np.percentile(write_times, 95) < 20, "FAIL: P95 write time exceeds 20ms"
```

```
print(" ✓ PASS\n")
```

```
print("=== All Tests Passed ===")
```

```
if __name__ == "__main__":
```

```
 benchmark()
```

```
...
```

```

```

## # 9. RISK ASSESSMENT

### ## 9.1 Technical Risks

Risk	Likelihood	Impact	Mitigation Strategy	Contingency Plan
TFLite accuracy < 75% in field	Medium (30%)	High	• Collect real-world dataset during pilot (500 images) • Retrain weekly during first month • A/B test model versions	Manual classification mode (user selects category, AI shows examples)
Raspberry Pi overheating in summer	High (60%)	Medium	• Thermal testing at 50°C ambient • Heatsink + fan combo • CPU throttling at 75°C	Reduce inference frequency (every 2nd frame), switch to Jetson Nano (costlier but better thermals)
SD card corruption from power cycles	Medium (25%)	High	• Industrial SLC SD cards • UPS battery (30s backup) • Read-only root filesystem (OverlayFS)	Automatic recovery mode (reinstall OS from recovery partition)
Touchscreen calibration drift	Low (10%)	Medium	• Auto-calibration routine weekly • Capacitive (not resistive) technology	Physical button navigation fallback
WiFi sync failures isolating terminals	Medium (35%)	Low	• 30-day local storage buffer • Retry with exponential backoff • Queue unsent data	Manual USB data extraction via service port
Camera lens dirt/obstruction	High (70%)	Medium	• Weekly cleaning schedule in maintenance plan • Image quality checks (detect low contrast)	Alert on maintenance dashboard, degrade gracefully to lower confidence threshold

---

## 9.2 Operational Risks

Risk	Likelihood	Impact	Mitigation	Contingency
Vandalism (screen damage)	Medium (40%)	High	• Gorilla Glass touchscreen • Tamper-evident enclosure • Surveillance camera integration	Insurance coverage, 24h replacement SLA
Theft of components	Low (15%)	High	• Secure mounting (theft-proof bolts) • GPS tracking module • Low resale value of specialized hardware	Asset tracking, police report triggers replacement
Power outages	High (50%)	Low	• UPS with 30-minute runtime • Graceful shutdown via NUT daemon	Resume operation automatically when power restored
User confusion/frustration	Medium (45%)	Medium	• 10-second tutorial on first use • Visual guides and animations • Multilingual support	On-site attendant during first 2 weeks, collect feedback via QR code survey
Seasonal usage drop (winter)	High (80%)	Low	• Heated enclosure (-20°C rating) • Encourage usage with "winter challenges"	Accept reduced usage, focus on summer/fall peak seasons

---

## 9.3 Project Delivery Risks

Risk	Likelihood	Impact	Mitigation	Contingency
72-hour deadline missed	Medium (35%)	High	• Prioritize MVP features (classification + basic UI)  • Use pre-trained models (no custom training)  • LLM-generated boilerplate code	Deploy v0.5 with manual classification, add AI in v1.0 (week 2)
Dataset labeling bottleneck	High (60%)	Medium	• Use TrashNet public dataset (2400 images) • Transfer learning reduces data needs (500 images sufficient)	Start with pre-trained model, refine with local data post-launch
Hardware delivery delays	Medium (30%)	High	• Order hardware in parallel with development • Prototype on standard RPi4 desktop setup • Use USB webcam for initial testing	Cloud-based demo (upload photo via web interface) until hardware arrives
Integration bugs (camera + TFLite)	Low (20%)	Medium	• Use proven libraries (picamera2 + TFLite runtime)  • Extensive error handling • Mock camera service for testing	Fallback to OpenCV VideoCapture (slower but reliable)

---

## 9.4 Risk Matrix

...

Impact vs Likelihood Matrix:

High Impact	SD Card Corruption		TFLite Accuracy
Medium Impact		Overheating Camera Dirt	Vandalism
Low Impact		WiFi Sync Failures	Seasonal Usage Drop
	Low (0-30%)	Medium (30-60%)	High (60%+)
	LIKELIHOOD		

Focus Areas (High Impact × High Likelihood):

- 1. None in critical quadrant - good risk profile
- 2. Medium risks require active monitoring

...

---

## 9.5 Risk Monitoring Dashboard

```
python
Risk indicators to track in production
risk_metrics = {
 "accuracy_degradation": {
 "metric": "7_day_moving_avg_confidence",
 "threshold": 0.70,
 "alert": "Retrain model with recent data"
 },
 "thermal_throttling": {
 "metric": "cpu_temp_above_75c_minutes_per_day",
```



```

 "threshold": 60, # More than 1hr/day is concerning
 "alert": "Upgrade cooling system"
 },
 "sync_failures": {
 "metric": "consecutive_failed_syncs",
 "threshold": 3,
 "alert": "Check network connectivity"
 },
 "user_abandonment": {
 "metric": "incomplete_interactions_percentage",
 "threshold": 0.20, # 20% abandonment
 "alert": "UX improvements needed"
 }
}
...

```

---

## # IMPLEMENTATION ROADMAP (72-Hour Sprint)

### ## Hour 0-8: Foundation

- [x] Initialize Git repository
- [x] Set up Raspberry Pi OS with SSH
- [x] Install Python 3.11, Node.js 20, dependencies
- [x] Create SQLite schema and seed data
- [x] Download and test TFLite model (pretrained MobileNetV3)

### ## Hour 8-24: Backend

- [x] FastAPI server with `/classify` endpoint
- [x] Camera service integration (picamera2)
- [x] TFLite inference pipeline
- [x] WebSocket live updates
- [x] Database logging
- [x] LLM-generated educational facts (150 items)

### ## Hour 24-40: Frontend

- [x] React app scaffolding (Vite + TypeScript)
- [x] 7 core components (LLM-generated)
- [x] WebSocket client integration
- [x] Framer Motion animations
- [x] TailwindCSS styling (high-contrast theme)

### ## Hour 40-56: Integration

- [x] End-to-end classification flow testing
- [x] Error handling implementation
- [x] Performance optimization (lazy loading, memoization)
- [x] Chromium kiosk mode configuration
- [x] systemd service setup

### ## Hour 56-72: Validation & Deployment

- [x] Field testing with sample waste items
- [x] Performance benchmarking (meets <800ms target)
- [x] Documentation (deployment guide)
- [x] Docker container (optional reproducibility)
- [x] Handoff presentation with live demo

---

## # APPENDIX: EXAMPLE PROMPTS FOR SPECIFIC TASKS

### ## A. Generating the Idle Screensaver Component

...

Create a React component for an idle screensaver that:

1. Displays rotating environmental facts every 8 seconds
2. Shows real-time daily recycling statistics (animated counters)
3. Has a "touch to wake" overlay
4. Uses Framer Motion for smooth transitions
5. Fetches data from WebSocket endpoint /ws/live

Technical requirements:

- TypeScript with strict mode
- TailwindCSS for styling (green/blue gradient background)
- Responsive to 1280x800 resolution
- Large touch-friendly fonts (minimum 24px)
- Accessibility: ARIA live region for screen readers

Include:

- Full component code with comments
- PropTypes interface
- Example usage
- CSS animations for counter increments

...

### ## B. Generating the Database Export Function

...

Write a Python function to export SQLite data to JSON for cloud sync:

Requirements:

- Export yesterday's classifications (full detail)
- Aggregate hourly statistics
- Calculate environmental impact using these coefficients:
  - \* Plastic: 1.2 kg CO<sub>2</sub>/item, 15 L water/item
  - \* Glass: 0.8 kg CO<sub>2</sub>/item, 8 L water/item
  - \* Paper: 0.5 kg CO<sub>2</sub>/item, 12 L water/item
- Handle missing data gracefully
- Compress output with gzip
- Return file path and metadata

Function signature:

```
def export_daily_stats(db_path: str, date: str) -> dict:
 """
```

Export daily statistics to JSON file

Args:

db\_path: Path to SQLite database  
date: Date to export (YYYY-MM-DD format)

Returns:

```
{
 "file_path": "/path/to/export.json.gz",
 "size_bytes": 12345,
 "record_count": 247,
 "export_timestamp": "2024-11-17T03:00:00Z"
}
```

Include error handling for database locks and disk space checks.

---

# CONCLUSION

This technical specification provides a **production-ready blueprint** for deploying the EcoVision smart waste terminal within the 72-hour prototype constraint. Key design principles:

- 1. **Offline-First Architecture**: Zero dependency on cloud services for core functionality
- 2. **LLM-Optimized Development**: Web stack chosen for maximum code generation quality
- 3. **Edge AI Performance**: TFLite + MobileNetV3 achieves <800ms inference on embedded hardware
- 4. **Graceful Degradation**: Multi-layer fallbacks ensure 99% uptime despite edge cases
- 5. **Rapid Iteration**: Modular architecture allows post-deployment improvements

**Expected Outcome**: A functional prototype demonstrating 85%+ classification accuracy, <10s user interaction time, and measurable impact on Kirov's recycling rates.

**Next Steps**:

- 1. Procure hardware (Est. \$250/unit × 5 pilot units = \$1,250)
- 2. Execute 72-hour development sprint using provided prompts
- 3. Deploy at 3 high-traffic locations (train station, university, shopping mall)
- 4. Collect 30 days of usage data for model refinement
- 5. Publish open-source repository for municipal replication

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